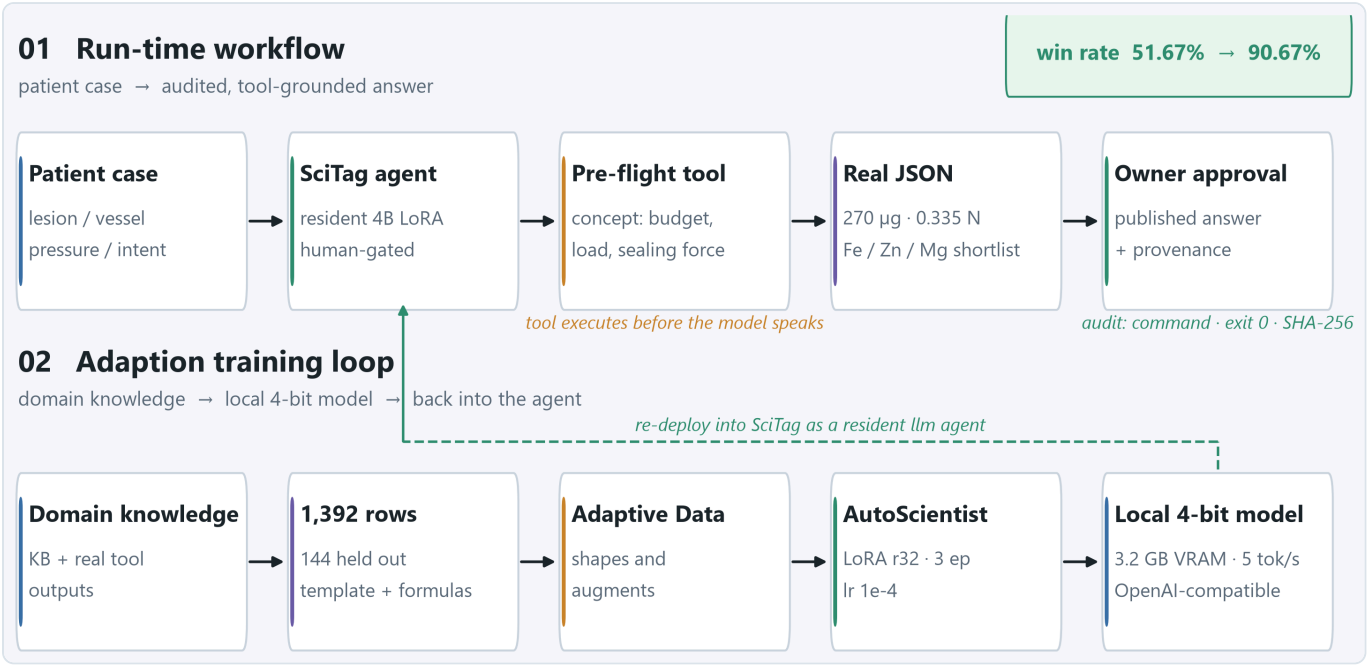
System direction — workflow, results, next steps

SciTag human-gated multi-agent harness × Adaption-tuned Qwen3.5-4B · system companion to the materials section



Approach

SciTag is a human-gated multi-agent harness: each agent mounts a domain package (knowledge base, skills, data, tools declared in *domain.json*). In the *vascular-stent* package, a zero-dependency Python tool computes drug-payload budgets, defect pressure loads, required sealing forces, and material shortlists with degradation screening; every run returns provenance (command, parameters, exit code, SHA-256 of data and script), and the agent reports to its human owner before publication.

Adaption turns the same knowledge into a local model: 1,392 instruction rows (144 held out) → Adaptive Data → AutoScientist LoRA on Qwen3.5-4B. The adapter is merged into the base weights and served 4-bit (≈3.2 GB VRAM, ≈5 tokens/s) behind an OpenAI-compatible endpoint, then registered in SciTag as a resident agent. The harness pre-executes the *concept* tool on the case text and feeds the real JSON to the model, so answers cite tool numbers rather than guesses.

What is new

(1) Tool-grounded auditing: the harness pre-executes the domain tools and records *analysisRuns* + provenance (command, exit code, SHA-256), so numbers are checkable rather than prompt-grounded. (2) A quantified training-strength ablation: same data, 51.67% → 90.67%, with a documented negative result for reasoning-trace data. (3) An end-to-end Adaption → local-agent path: merged LoRA, 4-bit, 3.2 GB VRAM, mounted inside a human-gated harness. (4) A case-driven material screen (*fdo*) returning per-polymer physical indicators (t_crit, mass loss, orifice drift, release order) alongside the concept decision, with explicit screening guardrails.

Results

Same data, three training runs: default LoRA **51.67%** win rate (tie with base), a reasoning-trace dataset **40.43%** (negative result), and stronger LoRA (r=32, 3 epochs, lr 1e-4) **90.67%**. On a local held-out check, numeric recall improved from **27.8% to 33.6%** and required-phrase coverage from **22.5% to 38.6%**. End to end, a hepatic-artery case (4.0 mm lesion, 3.0 mm vessel, 100 mmHg, permanent embolization) reports the drug arm at ≈270 µg (1.39×10^{-5} of an oral dose, not viable), the occlusion arm at 167 mN static load → **0.335 N** sealing force, and Fe/Zn/Mg frames passing the 1.5× margin — with an audit record (*concept* + *fdo*, exit 0, SHA-256). The harness passes its full suite (119 regression tests plus 10 tool-protocol tests).

Next steps

Add corrosion/fatigue kinetics and more polymer grades; collect 20–30 clinician reviews and run a DPO preference pass; benchmark the tuned 4B model against a larger open model; package one-command local deployment with a user guide and demo video.